Biermann 2022 melanoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

30 samples (brain-met MBM + peripheral-met MPM, sn/sc) with private melanoma CNV: pre-Harmony same-sample fraction 0.985. Harmony(θ=10) collapses it to one continuous component (99.97% of cells; 169 detached specks removed), mixing iLISI 1.0→6.4/30, met-sites interleave (1.78/2). Every one of 28 patients traverses the shared embryonic axis end-to-end (median span 94%) — a patient-SHARED continuous gradient, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

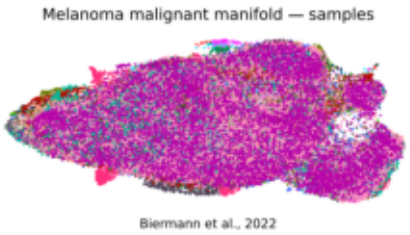
LOW-pSom, Crest-dominant melanoma (raw means Gast −0.05, pSom −0.02, Crest +0.05 — Crest the only positive module). No Gast/pSom co-expression hub (joint-high enrichment 0.91) — the two cross-repress in space (r=−0.57). Crest is an inverse TERMINAL gradient: depleted where pSom/Gast are high (joint 0.23 / 0.29), peaking at the trajectory terminus (Crest decile pt 0.60, latest; ρ=+0.32). Crest peaks at the terminus, not a Gast arm — no inversion flag.

Gate 3 — cross-repression → NA

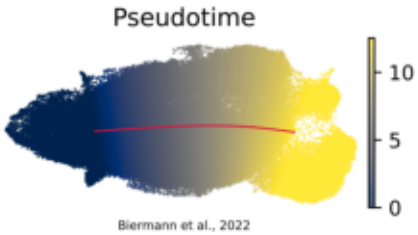
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks, fold-free; a PPT tree yields only spurious forks) and no Gast/pSom co-expression Mixed hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

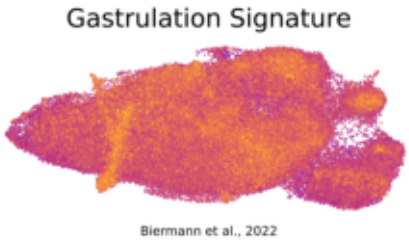
Manifold — samples



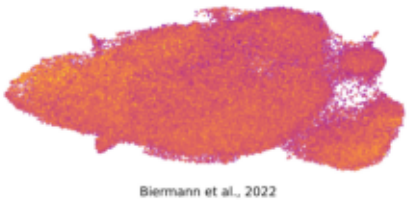
Pseudotime



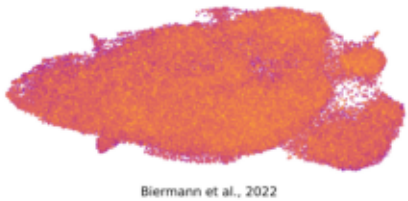
Gastrulation



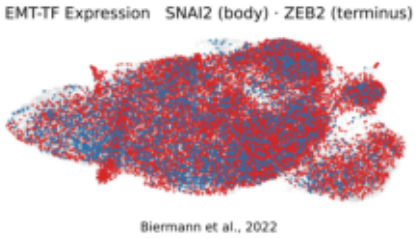
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic melanoma (Biermann et al., 2022), 10x — 30 samples (brain-met MBM + peripheral-met MPM, single-nucleus/single-cell); 136,973 cells master scope, 91,525 malignant; ≥100-cell floor keeps all 30 samples, 169 detached specks removed → 91,356 on the trajectory.

Verdict.
Non-branching pattern reproduces — G1 GO, G2 GO, G3 NA. The malignant manifold is a single monotone axis rooted at the Gast/pSom-high origin → terminal neural-crest end (pSom ρ=−0.50, Crest ρ=+0.32 along pseudotime); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The atlas's LOW-pSom, Crest-dominant case — Crest the only positive module, the neural-crest identity of a neural-crest-derived tumour. Cell-cycle confound checked: pSom tracks proliferation (r=+0.50) but Crest, the trajectory-defining module, is decoupled (−0.33), so the axis is developmental and the whole manifold is kept. EMT terminus-loaded: ZEB2 marks the neural-crest terminus (79%), SNAI2 the body (14%).

Parameter choices

Cells / patients	91,356 / 30 (≥100 floor)	**Integration**	Harmony θ=10; PCA 20PC; spread 1.3
Island removed	169 cells (0.18%, sn micro-blobs)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	SNAI2 (body) · ZEB2 (terminus); terminus-loaded		